

OMAR GARRAOUI

Milan, Italy

ogarraoui@icloud.com — omargarraoui.com — github.com/omargarraoui

EDUCATION

B.Sc. Mathematical Engineering (Applied Mathematics), Politecnico di Milano Sep 2025 – Jun 2028
Collegio di Milano – Competitive merit fellowship, full scholarship *Milan, Italy*

Coursework: Mathematical Analysis I & II, Linear Algebra and Geometry, Fundamentals of Computer Science

High School Diploma, ITST J.F. Kennedy Sep 2020 – Jun 2025
GPA: 3.9/4.0 Curriculum: Computer Science, Programming, Telecommunications *Pordenone, Italy*

HONORS & COMPETITIONS

Italian Mathematics Olympiad (2024) – National Semifinals qualifier; team captain.

ITACPC (Italian Collegiate Programming Contest) – 3rd place, competing individually in a team contest.

Collegio di Milano Fellowship (2025) · **Fratelli Zilli Scholarship** (A.Y. 2023/24, 2024/25)

PROJECTS

Contradiction Tensor [\[GitHub\]](#)

A decision-making tool that scores every pair of beliefs for **logical contradiction** via a **local NLI model** into an $N \times N \times K$ tensor indexed by market regime, then computes the minimum belief set to drop for internal consistency via a **greedy 2-approximation of Minimum Vertex Cover**.

Prediction Market Arbitrage Scanner [\[GitHub\]](#)

Detects **risk-free arbitrage** across prediction market venues by pricing binary contracts under taker ($c = \text{ask}(1+r)$) and profit-deferred ($c = \text{ask}/(1 - (1 - \text{ask})r)$) fee regimes; flags when $c_{\text{YES}} + c_{\text{NO}} < 1$ and allocates capital as $s_i \propto 1/c_i$ to **equalize expected payout** across outcomes.

CausalPulse [\[demo\]](#)

Treats the global news stream as a mixture of K **unobserved causal forces** and recovers them via **StrADiff** (arXiv:2604.04973); surfaces which upstream force is driving apparently unrelated coverage across domains, where standard topic classification sees only surface similarity.

EXPERIENCE

Research Associate, Polimi Data Scientists December 2025 – Present

- Company-sponsored projects with industry partners (**BCG, Bloomberg, Amazon**).
- Design and implement **ML pipelines** on real industry datasets.

Academic Mentor – Mathematics & Physics, ITST Kennedy Sep 2023 – May 2025
Tutored students 1-on-1 in calculus, linear algebra, and physics; designed individualized curricula.

PUBLICATIONS

Optimizing Exact String Matching via Statistical Anchoring [\[arXiv:2601.03271\]](#) – Enhances Boyer-Moore-Horspool by anchoring first mismatch checks to $c^* = \arg \min_{c \in P} \Pr(c \mid \text{corpus})$; faster window rejection at no added shift-heuristic complexity.

What Is Intelligence? [\[GitHub\]](#) – Unifies seven accounts of intelligence (Spearman through Deutsch) under the *Generative Model Revision* framework: intelligence as capacity to build and revise compressed world models under uncertainty.

SKILLS

Technical	Python, NumPy, Pandas, PyTorch, scikit-learn, C, C++, Java, Swift, Git, $\text{\LaTeX}$
Mathematics	Real Analysis, Linear Algebra, Probability & Statistics, Number Theory, Combinatorics
Languages	Arabic (native), Italian (native), English (C1), French (B2)